IoT configurations

IoT configuration involve several steps:

- **Configure the device:** It includes connecting the device with supply and calibration of the sensors connected with the system.
- **Diagnose the device:** running routine diagnosis can help detect potential problems before they occur and causes any problem.
- **Apply maintenance:** it involves update of firmware and replacement of any part or fix it before it causes any downtime. Maintenance of a device after regular interval enhance its life and help reducing any downtime due to error.
- **Ensure proper authentication:** proper authentication helps protecting the device. Also the network connected to the IoT device must be secure to protect it from any bad sector which may cause data corrupt or reduce the communication speed also.
- **Monitoring IoT devices:** monitoring the real time data may help reducing errors and prevent failures.

Basic components of IoT

- Devices and Sensors.
- Cloud Computing.
- User Interface.
- Networking connection.
- Gateway.

- Devices and sensors
- Devices and sensors are the components of the connectivity layer. These smart sensors continuously collect data from the environment and transmit it to the next layer. For example, our phone is a device with multiple sensors such as GPS, camera, motion, etc.
- The latest techniques in semiconductor technology enable intelligent micro-sensors to be produced for various applications.
- There are some common sensors are as follows:
  - Temperature and thermostats sensors
  - Pressure sensors
  - Humidity / Moisture level
  - Light intensity detectors
  - Moisture sensors
  - Proximity detection
  - RFID tags

Edge Computing

- To reduce the data consumption and latency issues occurred due to sending data in a centralised server (cloud server). The edge devices are used which are closer to the end user.
- The centralised hub shifts some of its storage resources and processing servers to different places so that can access fast by the users is known as edge computing.
- Usually all the operations can be performed at cloud server.
- Ede computing means running fewer processes in the cloud and moving the processes to the local servers.
- The edge is an intermediator node between user and cloud server.

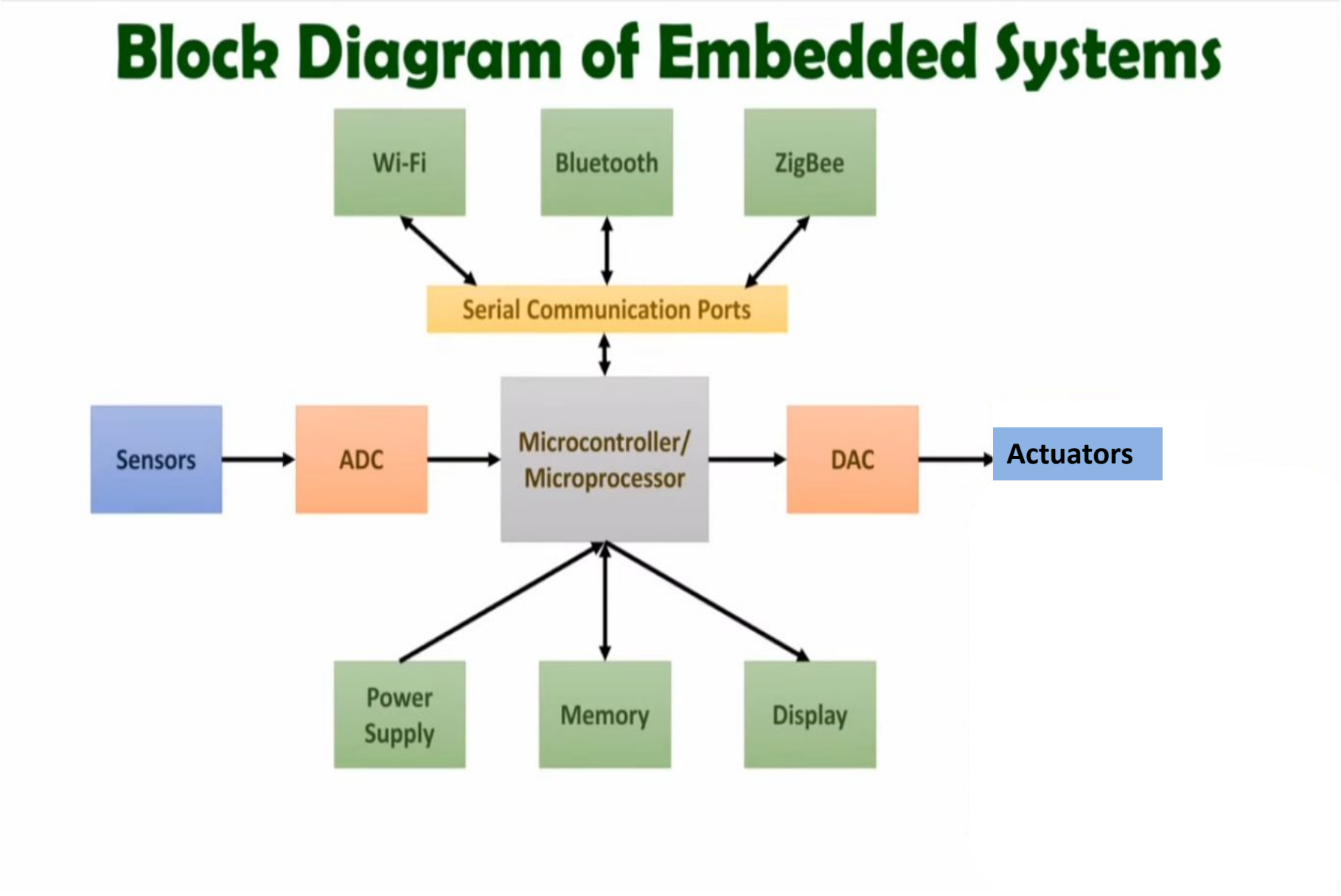
Benefits of Edge computing

- Reduced usage of cloud resources.
- Faster and real time response. (with reduced bandwidth and low latency)
- No communication gap.

Drawbacks of Edge Computing:

- Additional resource is required.

Embedded systems in IoT



- An embedded system is a computer system that has computer hardware and software embedded to perform specific tasks.
- In contrast to general purpose computers which can perform various types of tasks, embedded system is designed to perform only specific type of tasks.
- Some embedded systems have specialized processors such as Digital signal processors, graphics processors, and application-specific processors.
- Embedded systems usually runs embedded operating systems for eg., real time operating systems (RTOS).